

Dongwook Kwon

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Experience

Suresoft Technologies

Seongnam-si, Korea

AI ENGINEER, INTELLIGENT AGENT TECHNOLOGY TEAM

Dec. 2025 - Present

- Developing an AI validation system utilizing AI agent-based architectures for enterprise model verification.
- Designing autonomous and collaborative AI agents that verify and evaluate machine-learning models at scale.
- Engineered and delivered a critical agent-based codebase for a mission-critical vehicle control data project for Hyundai Motor Company.

Bio-Computing and Machine Learning Laboratory, Kwangwoon University

Seoul, Korea

RESEARCH ASSISTANT

Aug. 2021 - Dec. 2025

- Researched deep-learning algorithms for anomaly detection in complex time-series data, focusing on critical cyber-physical systems including nuclear power generation environments.
- Co-developed an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.

LG Electronics Canada (with University of Toronto)

Toronto, ON, Canada

VISITING RESEARCHER

Jan. 2025 - Jul. 2025

- Built an LLM-driven assessment framework for evaluating performance of agentic AI systems.
- Work fed directly into peer-reviewed publications at EMNLP 2025 and ACL 2026.

Qualcomm Institute, University of California San Diego

San Diego, CA, USA

AI DEVELOPMENT PROJECT PARTICIPANT, SUMMER PROGRAM

Jul. 2022 - Aug. 2022

- Built a personality-based drug-addiction prediction system using machine-learning models.
- Recognized with the program's Achievement Award.

Education

Kwangwoon University

Seoul, Korea

MASTER OF SCIENCE IN COMPUTER ENGINEERING

Mar. 2024 - Feb. 2026

- GPA: 4.33/4.5 (98.1%)

University of Toronto

Toronto, ON, Canada

GRADUATE RESEARCH PROGRAM IN MECHANICAL AND INDUSTRIAL ENGINEERING

Jan. 2025 - Jul. 2025

- GPA: 3.68/4.0

Kwangwoon University

Seoul, Korea

BACHELOR OF SCIENCE IN ELECTRONICS AND COMMUNICATIONS ENGINEERING

Mar. 2019 - Feb. 2024

- GPA: 3.41/4.5 (88.1%)

Honors & Awards

MAJOR AWARDS

Aug. 2022 **Achievement Award**, Qualcomm Institute AI Development Project — UC San Diego

United States

Dec. 2021 **Gold Award (Ministerial Award)**, 2021 Smart Maritime Logistics Project — Minister of Oceans and Fisheries

South Korea

ACADEMIC RECOGNITION

Nov. 2025 **Outstanding Student Paper Award**, Conference of the Institute of Electronics and Information Engineers

South Korea

Nov. 2024 **Best Poster Award (Silver)**, 10th International Biomedical Engineering Conference (IBEC 2024)

South Korea

Nov. 2023 **Outstanding Paper Award**, Conference of the Korean Society of Medical and Biological Engineering

South Korea

COMPETITIONS

Dec. 2025 **Silver Award**, 23rd Embedded Software Competition (KIISE)

South Korea

Dec. 2022 **Bronze Award**, Hanium ICT Mentoring Competition — FKII

South Korea

Sep. 2022 **Excellence Award**, 18th Kwangwoon ICT Exhibition (KWIX) — Kwangwoon University

South Korea

Oct. 2021 **Grand Prize**, MY (Multi-Y) Capstone Design Competition — Kwangwoon University

South Korea

Funded Research Projects

LG Electronics Canada

AGENTIC AI EVALUATION FRAMEWORK

- Developed an evaluation framework for agentic AI systems using LLMs to assess performance.

Sejong University

AI-BASED INTRUSION DETECTION AND ATTACK PACKET CRAFTING TECHNOLOGIES FOR APR1400

- Developed an AI defense system to protect nuclear power plants from cyber attacks.

Ministry of Science and ICT

QUADRUPEL ROBOT USING VISION SLAM WITH A DEPTH CAMERA

- Developed an autonomous quadruped robot using a navigation algorithm with SLAM and a depth camera.

Qualcomm Institute, University of California San Diego

PERSONALITY-BASED DRUG ADDICTION PREDICTION SYSTEM

- Developed a personality-based drug addiction prediction system using a machine learning model.

Kookmin University

DEVELOPMENT OF ACTIVE KILL-SWITCH AND BIOMARKER-BASED DEFENSE SYSTEM FOR LIFE-THREATENING IOT MEDICAL DEVICES

- Developed a security system for heart implant devices using an anomaly detection algorithm.

Ministry of Oceans and Fisheries

DEVELOPMENT OF A SMART LOGISTICS SYSTEM USING COMPUTER VISION

- Developed a logistics detection system incorporating location tracking and stack-layer identification for containers.

Kwangwoon University

SMART KITCHEN IOT SYSTEM UTILIZING COMPUTER VISION AND DEEP LEARNING

- Developed a cloud IoT platform featuring a system for classifying groceries in the fridge and an OCR receipt program.

Toronto, ON, Canada

Jan. 2025 - Present

Seoul, Korea

Jul. 2022 - Present

Seoul, Korea

Apr. 2022 - Nov. 2022

San Diego, CA, USA

Jul. 2022 - Aug. 2022

Seoul, Korea

Sep. 2021 - Dec. 2022

Seoul, Korea

Apr. 2021 - Nov. 2021

Seoul, Korea

May 2021 - Oct. 2021

Publications

JOURNAL ARTICLES

- Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park, "Designing a remote photoplethysmography-based heart rate estimation algorithm during a treadmill exercise," *Electronics*, vol. 14, no. 5, p. 890, 2025.
- **D. Kwon**, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park, "Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments," under review at *Data Mining and Knowledge Discovery (Springer Nature)*, 2025.

CONFERENCE PAPERS

- R. Chang[†], **D. Kwon**[†], J. Lee[†], and N. Verma, "CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades," in *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics (ACL) — Industry Track (Poster)*, 2026.
[†]Equal contribution.
- J. Lee[†], R. Chang[†], **D. Kwon**[†], H. Singh, and N. Verma, "GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems," in *Proceedings of the EMNLP 2025 Industry Track (Oral)*, 2025.
[†]Equal contribution.
- **D. Kwon**, M. Kim, Y. Kang, J. Lee, and C. Park, "Hybrid neural network model for anomaly detection in implantable devices using graph attention networks and transformers," in *Proc. 10th Int. Biomed. Eng. Conf. (IBEC)*, 2024. **Best Poster Award — Silver.**
- **D. Kwon**, Y. Kang, and C. Park, "Enhancing medical device security with GNN-GRU anomaly detection model," in *Proc. 62nd KOSOMBE Autumn Conf.*, pp. 204–205, 2023. **Outstanding Paper Award.**
- J. Yang, J. Kim, J. Lee, H. Ryu, **D. Kwon**, S. Yeo, P. Kim, Y. Kim, J. Lim, H. Yoon, and C. Park, "Metaverse: Design of the Car Price Prediction Model Through a Machine-learning Approach," in *Proc. IEEE Int. Conf. Metaverse Computing, Networking and Applications (MetaCom)*, pp. 734–737, 2023.

Patents

REGISTERED

Apr 2026 **Method for Implantable Medical Device to Detect Anomaly in Real Time**, KR 10-2961984

Republic of Korea

Aug 2025 **GPT API-Based Network Anomaly Detection**, KR 10-2848814

Republic of Korea

PENDING APPLICATIONS

Oct 2025 **System for Anomaly Detection Using a Collaborative Multi-AI Determination Module Based on LLM**, KR
10-2025-0160217
Aug 2024 **Categorical Data of Networks Based Network Anomaly Detection Apparatus and Method Thereof**, KR
10-2024-0110559

Republic of Korea

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Leadership Experience

Kwangwoon International Student Association

PRESIDENT

- Lead intercultural programs and student engagement initiatives.

Kwangwoon Broadcasting Center (KWBC)

DIRECTOR OF FILMING AND EDITING

- Oversaw media production and content delivery.

Dept. of Electronics and Communications Engineering

CLASS REPRESENTATIVE

- Represented student interests in academic affairs.

Amateur Radio Club

EXECUTIVE MEMBER

- Organized radio workshops and technical exhibitions.